

PocketLab BLE Design

A durable reference for the function generator's BLE architecture, safety model, and extensible GATT namespace.

SYSTEM AT A GLANCE

MOBILE APP

React Native / Expo

BLE LINK

Custom GATT service

CONTROL

ESP32 + NimBLE

SIGNAL

AD9833 DDS

DESIGN BASELINE

Control objective

Low-latency, bidirectional control with explicit device state, predictable disconnect behavior, and room for future PocketLab instruments.

Logical safe idle

WAVE=DC FREQ=0
AMP=0.00 OFFSET=0.00
OUTPUT=OFF



DC is an internal safe-state sentinel. The AD9833 is not asked to synthesize DC; hardware output is disabled instead.

GATT STRUCTURE AND UUID STRATEGY

SUFFIX	ATTRIBUTE	ACCESS	ROLE	STATUS
0001	PocketLab service	Primary service	Function-generator control namespace	In use
0002	Command	Write / Write NR	App -> ESP32 request frames	In use
0003	Response	Notify + Read	ESP32 -> app ACK, ERR, data	In use
0004	State	Read + Notify	Snapshot and unsolicited state changes	Reserved
0005	Device info	Read	Protocol version and capabilities	Reserved

SERVICE 8f5b0001-6c4d-4a73-a8f1-3d9ea01c0001

COMMAND 8f5b0002-6c4d-4a73-a8f1-3d9ea01c0001

Strategy: replace N in 8f5b000N with the table suffix; keep the shared 128-bit base for every function-generator attribute.

NimBLE Migration & Debugging

The important failures, observations, and design rules preserved as reusable project knowledge.

WHY NIMBLE

Lean stack

Lower memory overhead and a smaller embedded BLE surface for the ESP32.

Explicit behavior

Characteristic properties, callbacks, notifications, and advertising are configured deliberately.

Stable foundation

A consistent server API for reconnect handling and future PocketLab services.

DEBUGGING HISTORY

10 JUL Generic GATT baseline

BAAD/F00D demo reads worked. Writes appeared successful on Android, but the ESP32 onWrite callback did not run and the original value remained unchanged.

11 JUL Write-path isolation

Request and Command modes plus write-with-response and write-without-response were tested. In one nRF Connect test, the attribute value changed to PING while callback logging was still absent.

11-12 JUL Custom service + reconnect

Service discovery, characteristic reads, notifications, connection status, and automatic reconnect worked. A PING write was accepted, then the app timed out after 5 seconds waiting for a response.

12 JUL NimBLE command path operational

The custom command server began delivering responsive app control. Fast round trips confirmed the BLE path was viable for interactive waveform control.

13 JUL Safe-idle hardening

Disconnect behavior exposed an unwanted roughly 3.3 V DC output. The design separated logical DC idle from supported AD9833 waveforms and made OUTPUT=OFF the default and disconnect action.

MIGRATION GUARDRAILS

AREA	RULE TO KEEP
Callbacks	Attach characteristic callbacks before starting the service; log payload length and decoded text at entry.
Properties	Firmware properties must match the app's selected write type. Support both only when both paths are tested.
Notifications	Subscribe before sending the first command; create one response for every accepted request.
Reconnect	On disconnect, force safe idle, restart advertising, and let only the app initiate automatic reconnect.
State	Keep logical state separate from hardware modes; DC is never passed to the AD9833 waveform function.

PocketLab Command Protocol v1

A compact ASCII request/response protocol designed to remain reliable at the default BLE MTU of 23 bytes.

WIRE FORMAT

REQUEST

SET:FREQ:1000

SUCCESS

OK:FREQ:1000

FAILURE

ERR:RANGE:FREQ

UTF-8/ASCII, one frame per characteristic write or notification, no terminator required. Keep v1 frames at 20 bytes or fewer so they fit the MTU-3 payload without fragmentation. The app permits one outstanding request at a time.

COMMAND SET

REQUEST	PURPOSE	SUCCESS RESPONSE
PING	Verify the complete write/notify path	OK:PONG
GET:STATE	Request a complete state transaction	STATE:* frames, then STATE:END
SET:WAVE:SINE	Set SINE, TRIANGLE, or SQUARE	OK:WAVE:SINE
SET:FREQ:1000	Set output frequency in hertz	OK:FREQ:1000
SET:AMP:0.65	Set requested amplitude in Vpp	OK:AMP:0.65
SET:OFFSET:0.00	Set requested DC offset in volts	OK:OFFSET:0.00
OUTPUT:ON	Enable hardware output after validation	OK:OUTPUT:ON
OUTPUT:OFF	Immediately disable hardware output	OK:OUTPUT:OFF

STATE TRANSACTION

STATE:WAVE:SINE

STATE:FREQ:1000

STATE:AMP:0.65

STATE:OFFSET:0.00

STATE:OUTPUT:OFF

STATE:END

VALIDATION AND FAILURE BEHAVIOR

Firmware invariants

- Reject malformed, unsupported, or out-of-range values.
- Return exactly one OK or ERR for every accepted request.
- Apply OUTPUT:OFF before disconnect cleanup or re-advertising.
- Never map the DC sentinel into an AD9833 waveform call.

Definition of done Every command is observable, acknowledged, range-checked, and safe across disconnect/reconnect.

Implementation sequence

- 1 Add parser + unit tests
- 2 Add response notification queue
- 3 Serialize app requests + timeout
- 4 Verify MTU 23 and reconnect cases
- 5 Record scope readings by waveform